

Until now we have discussed plane curves. For curves in space, we need a few major modifications.

Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3$$

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Again we have a parametrization but this time the target is the 3-dimensional space

Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3$$



Spatial curves

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Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3$$

$$\theta(t) = (x(t), y(t), z(t))$$

○

and so it has 3 coordinates in the target instead of just two

Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3$$

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Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3$$

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Example

Let us consider one example of a spatial curve

Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3$$

$$\theta(t) = (x(t), y(t), z(t))$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$

Notice if we were to project this to the first two coordinates, we would get a circle

Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3$$

$$\theta(t) = (x(t), y(t), z(t))$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$

but as we increase t the z -coordinate increases

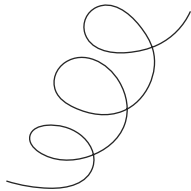
Spatial curves

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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



and we get a helix. In this picture, the axes, and therefore also the helix, is tilted.

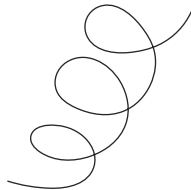
Spatial curves

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Spatial curves

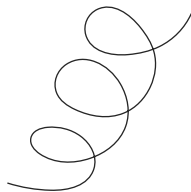
$$\theta : (a, b) \rightarrow \mathbb{R}^3$$

$$\theta(t) = (x(t), y(t), z(t))$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



As before, we can talk about the velocity vector of the curve at point $\theta(t)$

Spatial curves

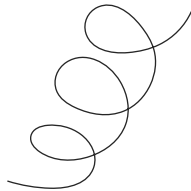
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Spatial curves

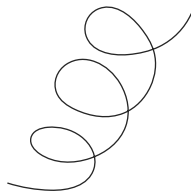
$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



and if we chose the parametrization to be unit speed

Spatial curves

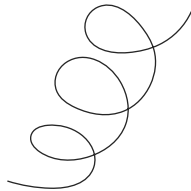
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Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

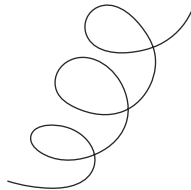
$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



As before, we call that unit vector, $\mathbf{T}(t)$

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

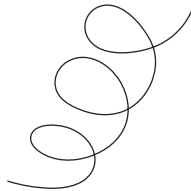
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Spatial curves

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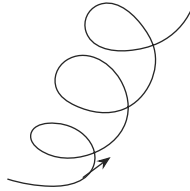
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Spatial curves

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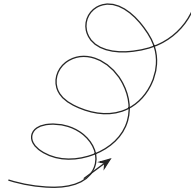
$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



The derivative of $\mathbf{T}(t)$ is precisely the acceleration vector

Spatial curves

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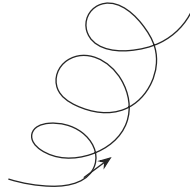
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



and if we divide by its magnitude we will get what we will call the normal vector.

Spatial curves

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$$\theta(t) = (x(t), y(t), z(t))$$

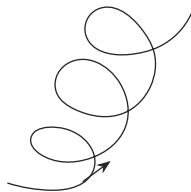
$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



Note that for curves in the plane we could define the normal vector by simply rotating the tangent by $\pi/2$

Spatial curves

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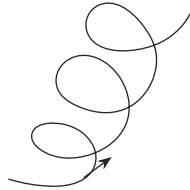
$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



but now there is no unique plane to rotate it in

Spatial curves

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$$\theta(t) = (x(t), y(t), z(t))$$

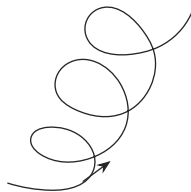
$$\mathbf{T}(t) =$$

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$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



Remember how useful $\mathbf{T}(t)$ and $\mathbf{N}(t)$ were in understanding plane curves!

Spatial curves

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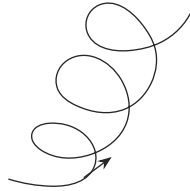
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



For instance, they allowed us to understand how $\mathbf{N}(t)$ was changing

Spatial curves

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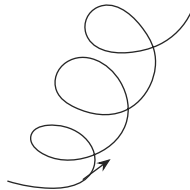
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



in terms of these two vectors so that we could identify the potential center of the best fit circle

Spatial curves

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$$\theta(t) = (x(t), y(t), z(t))$$

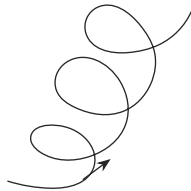
$$\mathbf{T}(t) =$$

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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



We will try to extend that useful notion to curve in space

Spatial curves

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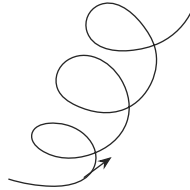
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



But as of now we have only two unit vectors that are perpendicular

Spatial curves

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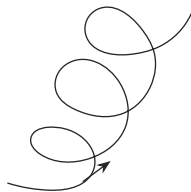
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



with them we can only define a plane worth of vectors, but we are now in a 3-dimensional space

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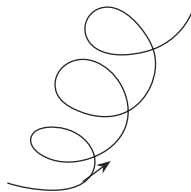
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Example

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We need a third unit vector so that all three are unit and perpendicular to each other

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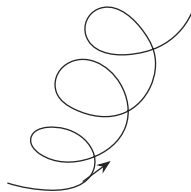
$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



The cross product is perfect for this!

Spatial curves

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$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

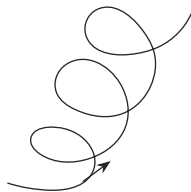
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$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



Since the original vectors are perpendicular *and* unit, so will the cross product

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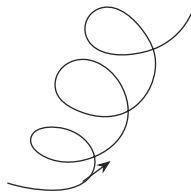
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Spatial curves

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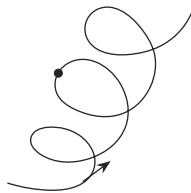
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



So if we have a point $\theta(t)$ as shown in the figure

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

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$$\mathbf{T}(t) =$$

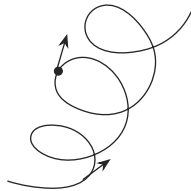
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



then we have its unit tangent vector

Spatial curves

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$$\mathbf{T}(t) =$$

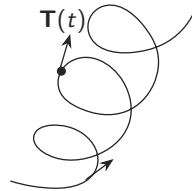
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



which we call $\mathbf{T}(t)$

Spatial curves

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$$\mathbf{T}(t) =$$

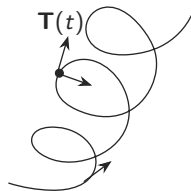
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

The unit vector in the direction of the acceleration gives us another which is perpendicular to $\mathbf{T}(t)$

Spatial curves

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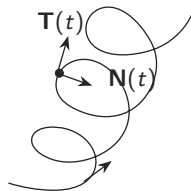
$$\theta'(t) = (x'(t), y'(t), z'(t))$$

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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

We call that $\mathbf{N}(t)$

Spatial curves

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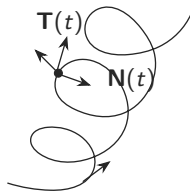
$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

while the cross product is the third unit vector

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

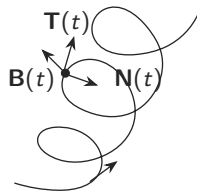
$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

which we call $\mathbb{B}(t)$

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

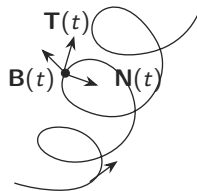
$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

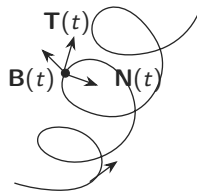
$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

As discussed, it would be useful to write the derivatives of each of these three vectors

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

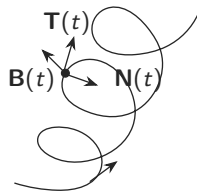
$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) =$$

$$\mathbf{N}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

in terms of the original three vectors

Spatial curves

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$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

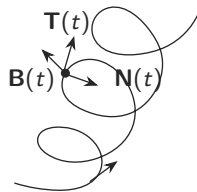
$$\mathbf{T}'(t) =$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

Notice that $\mathbf{N}(t)$ is defined to be in the direction of $\mathbf{T}'(t)$

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

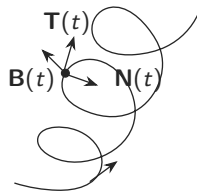
$$\mathbf{T}'(t) =$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

and, therefore, the other perpendicular vectors do not contribute to it

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

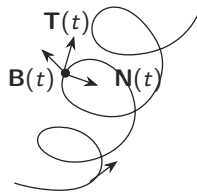
$$\mathbf{T}'(t) = \kappa \mathbf{N}(t) +$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

That is why it has no contribution from $\mathbf{T}(t)$

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

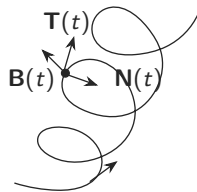
$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) +$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

while it is of course in the direction of the unit vector $\mathbf{N}(t)$ which has the curvature as magnitude

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

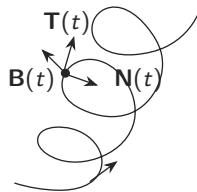
$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

and just as with $\mathbf{T}(t)$, there is no contribution from $\mathbf{B}(t)$ which is perpendicular to its direction

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

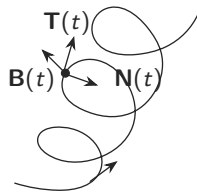
$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

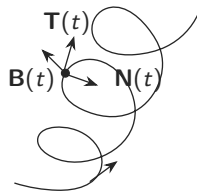
$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

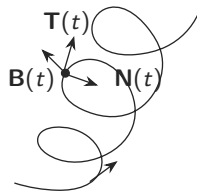
$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

To understand the others, let us understand this general procedure

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

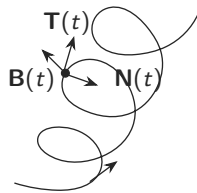
$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

where we have three vectors perpendicular to each other

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

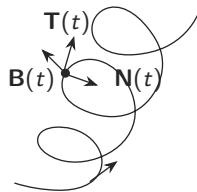
$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

Can any other vector can be written as some combination of them?

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

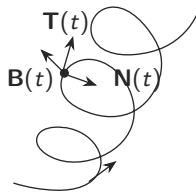
$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

and if so, what combination? In other words, what are the coefficients a , b , and c in the highlighted equation

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

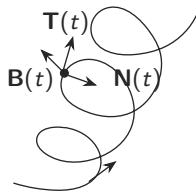
$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

We can exploit the fact that they are perpendicular by using the dot product in the following way

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

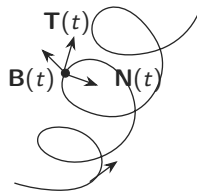
Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

If we take the dot product of the entire equation with $\mathbf{v}_1$

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

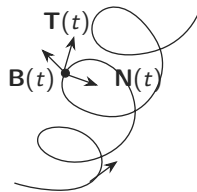
Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 +$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

then in the first summand we are simply getting a times the dot product of $\mathbf{v}_1$ with itself

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

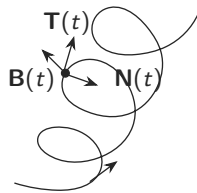
Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 +$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

but in the second summand notice that the dot product should be 0 because they are perpendicular

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

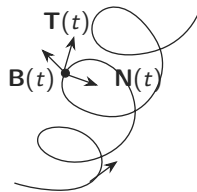
Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

and also with the the third summand

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

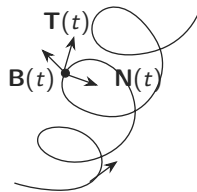
$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

this enables us to get an equation with a as the only unknown

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

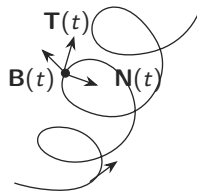
$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

In fact, we can simplify matters a bit

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

and unit vectors

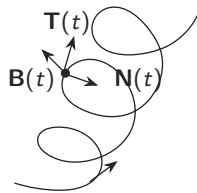
$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

by adding the condition that they are unit vectors, apart from being perpendicular

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other

and unit vectors

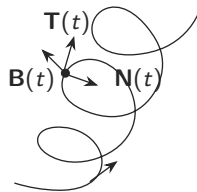
$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 =$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other
and unit vectors

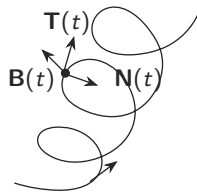
$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) +$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

and that would mean that the dot product of $\mathbf{v}_1$ with itself will be 1

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) =$$

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Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other
and unit vectors

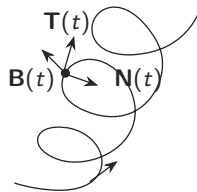
$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) +$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

and because the others are perpendicular

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

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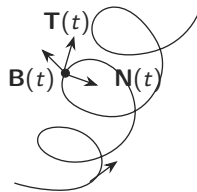
$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0$$

Example

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○

they are 0

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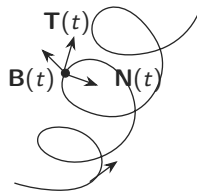
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Example

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○

and so we can easily figure out a

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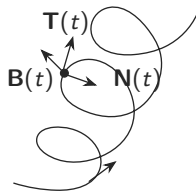
$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0 + 0$$

$$\text{So, } a = \mathbf{w} \cdot \mathbf{v}_1$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

as the dot product of $\mathbf{w}$ with $\mathbf{v}_1$

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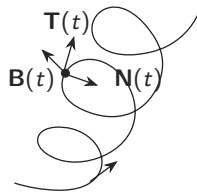
$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0 + 0$$

$$\text{So, } a = \mathbf{w} \cdot \mathbf{v}_1$$

Example

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○

if we wanted to figure out b we would have taken the dot product with $\mathbf{v}_2$ and so on

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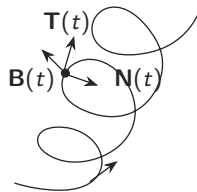
$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0 + 0$$

$$\text{So, } a = \mathbf{w} \cdot \mathbf{v}_1$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) = ?$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

So let us use this to figure out how to write $\mathbf{N}'(t)$

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$$\mathbf{N}'(t) =$$

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Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other and unit vectors

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$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0 + 0$$

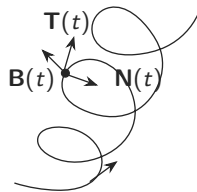
$$\text{So, } a = \mathbf{w} \cdot \mathbf{v}_1$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) = ?$$

$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

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$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0 + 0$$

$$\text{So, } a = \mathbf{w} \cdot \mathbf{v}_1$$

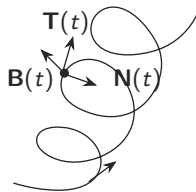
$$\mathbf{N}'(t) \cdot \mathbf{T}(t) = ?$$

$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



○

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$$\mathbf{N}'(t) \cdot \mathbf{T}(t) = ?$$

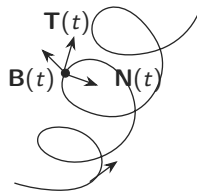
$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$

$$\text{So } \mathbf{N}'(t) \cdot \mathbf{T}(t) = -\mathbf{N}(t) \cdot \mathbf{T}'(t)$$

Example

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○

Spatial curves

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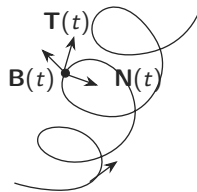
$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

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Example

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○

Spatial curves

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$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0 + 0$$

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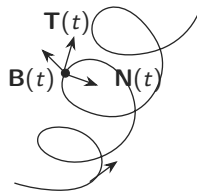
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Example

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○

Spatial curves

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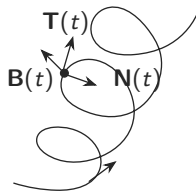
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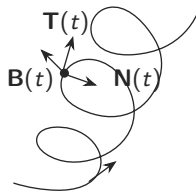
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Example

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○

Spatial curves

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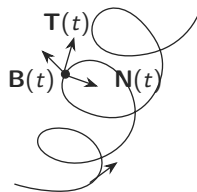
$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

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Example

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Spatial curves

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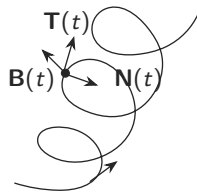
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Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



$$\mathbf{N}'(t) \cdot \mathbf{B}(t) = ?$$

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

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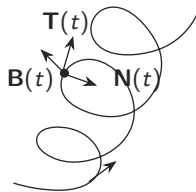
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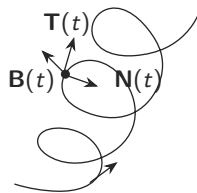
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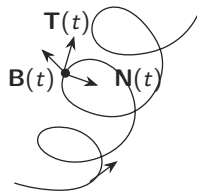
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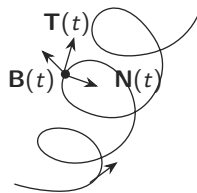
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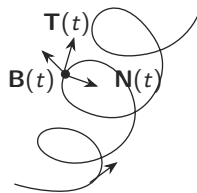
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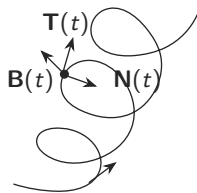
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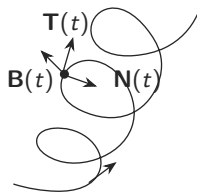
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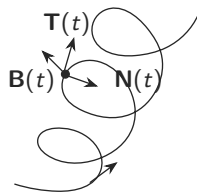
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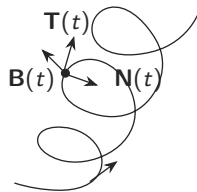
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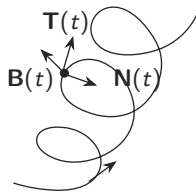
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$$\mathbf{N}'(t) \cdot \mathbf{T}(t) = ?$$

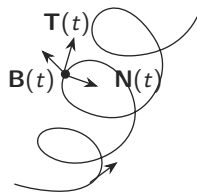
$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$

$$\text{So } \mathbf{N}'(t) \cdot \mathbf{T}(t) = -\mathbf{N}(t) \cdot \mathbf{T}'(t) = -\kappa(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



$$\mathbf{N}'(t) \cdot \mathbf{B}(t) = ?$$

$$\mathbf{N}(t) \cdot \mathbf{B}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{B}(t) = -\mathbf{N}(t) \cdot \mathbf{B}'(t) = \tau(t) \text{ (Torsion of } \theta \text{ at } \theta(t))$$

$$\mathbf{B}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{B}'(t) \cdot \mathbf{T}(t) + \mathbf{B}(t) \cdot \mathbf{T}'(t) = 0$$

$$\mathbf{B}'(t) \cdot \mathbf{T}(t) + \mathbf{B}(t) \cdot \kappa(t)\mathbf{N}(t) = 0$$

$$\mathbf{B}'(t) \cdot \mathbf{T}(t) = 0$$

Spatial curves

$$\theta : (a, b) \rightarrow \mathbb{R}^3 \text{ unit speed}$$

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) = -\kappa(t)\mathbf{T}(t) + 0\mathbf{N}(t) + \tau(t).\mathbf{B}(t)$$

$$\mathbf{B}'(t) =$$

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are perpendicular to each other and unit vectors

$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w}.\mathbf{v}_1 = a\mathbf{v}_1.\mathbf{v}_1 + b\mathbf{v}_2.\mathbf{v}_1 + c\mathbf{v}_3.\mathbf{v}_1$$

$$\mathbf{w}.\mathbf{v}_1 = a(1) + 0 + 0$$

$$\text{So, } a = \mathbf{w}.\mathbf{v}_1$$

$$\mathbf{N}'(t).\mathbf{T}(t) = ?$$

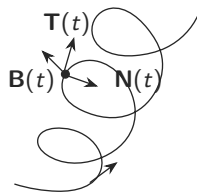
$$\mathbf{N}(t).\mathbf{T}(t) = 0$$

$$\mathbf{N}'(t).\mathbf{T}(t) + \mathbf{N}(t).\mathbf{T}'(t) = 0$$

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$$\theta(t) = (\cos(t), \sin(t), t)$$



$$\mathbf{N}'(t).\mathbf{B}(t) = ?$$

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$$\mathbf{B}(t).\mathbf{T}(t) = 0$$

$$\mathbf{B}'(t).\mathbf{T}(t) + \mathbf{B}(t).\mathbf{T}'(t) = 0$$

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Spatial curves

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$$\text{So, } a = \mathbf{w}.\mathbf{v}_1$$

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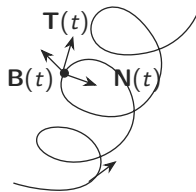
$$\mathbf{N}(t).\mathbf{T}(t) = 0$$

$$\mathbf{N}'(t).\mathbf{T}(t) + \mathbf{N}(t).\mathbf{T}'(t) = 0$$

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$$\mathbf{B}'(t).\mathbf{T}(t) + \mathbf{B}(t).\mathbf{T}'(t) = 0$$

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$$\mathbf{B}'(t) = 0\mathbf{T}(t) - \tau(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

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$$\mathbf{w}.\mathbf{v}_1 = a(1) + 0 + 0$$

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$$\mathbf{N}'(t).\mathbf{T}(t) = ?$$

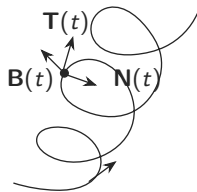
$$\mathbf{N}(t).\mathbf{T}(t) = 0$$

$$\mathbf{N}'(t).\mathbf{T}(t) + \mathbf{N}(t).\mathbf{T}'(t) = 0$$

$$\text{So } \mathbf{N}'(t).\mathbf{T}(t) = -\mathbf{N}(t).\mathbf{T}'(t) = -\kappa(t)$$

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$$\theta(t) = (\cos(t), \sin(t), t)$$



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$$\mathbf{N}'(t).\mathbf{B}(t) = -\mathbf{N}(t).\mathbf{B}'(t) = \tau(t) \text{ (Torsion of } \theta \text{ at } \theta(t))$$

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$$\mathbf{B}'(t).\mathbf{T}(t) + \mathbf{B}(t).\mathbf{T}'(t) = 0$$

$$\mathbf{B}'(t).\mathbf{T}(t) + \mathbf{B}(t).\kappa(t)\mathbf{N}(t) = 0$$

$$\mathbf{B}'(t).\mathbf{T}(t) = 0$$

Spatial curves

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$$\mathbf{N}'(t) = -\kappa(t)\mathbf{T}(t) + 0\mathbf{N}(t) + \tau(t)\mathbf{B}(t)$$

$$\mathbf{B}'(t) = 0\mathbf{T}(t) - \tau(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

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$$\mathbf{w} = a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a\mathbf{v}_1 \cdot \mathbf{v}_1 + b\mathbf{v}_2 \cdot \mathbf{v}_1 + c\mathbf{v}_3 \cdot \mathbf{v}_1$$

$$\mathbf{w} \cdot \mathbf{v}_1 = a(1) + 0 + 0$$

$$\text{So, } a = \mathbf{w} \cdot \mathbf{v}_1$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) = ?$$

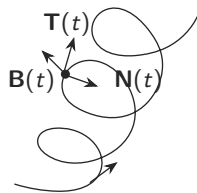
$$\mathbf{N}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{T}(t) + \mathbf{N}(t) \cdot \mathbf{T}'(t) = 0$$

$$\text{So } \mathbf{N}'(t) \cdot \mathbf{T}(t) = -\mathbf{N}(t) \cdot \mathbf{T}'(t) = -\kappa(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



$$\mathbf{N}'(t) \cdot \mathbf{B}(t) = ?$$

$$\mathbf{N}(t) \cdot \mathbf{B}(t) = 0$$

$$\mathbf{N}'(t) \cdot \mathbf{B}(t) = -\mathbf{N}(t) \cdot \mathbf{B}'(t) = \tau(t) \text{ (Torsion of } \theta \text{ at } \theta(t))$$

$$\mathbf{B}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{B}'(t) \cdot \mathbf{T}(t) + \mathbf{B}(t) \cdot \mathbf{T}'(t) = 0$$

$$\mathbf{B}'(t) \cdot \mathbf{T}(t) + \mathbf{B}(t) \cdot \kappa(t)\mathbf{N}(t) = 0$$

$$\mathbf{B}'(t) \cdot \mathbf{T}(t) = 0$$

Exercise: Show that if the torsion of a curve is constant 0, then the curve lies on a plane

Spatial curves

$\theta : (a, b) \rightarrow \mathbb{R}^3$ unit speed

$$\theta(t) = (x(t), y(t), z(t))$$

$$\mathbf{T}(t) =$$

$$\theta'(t) = (x'(t), y'(t), z'(t))$$

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$$

$$\mathbf{B}(t) = \mathbf{T}(t) \times \mathbf{N}(t)$$

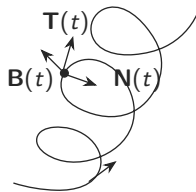
$$\mathbf{T}'(t) = 0\mathbf{T}(t) + \kappa(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

$$\mathbf{N}'(t) = -\kappa(t)\mathbf{T}(t) + 0\mathbf{N}(t) + \tau(t)\mathbf{B}(t)$$

$$\mathbf{B}'(t) = 0\mathbf{T}(t) - \tau(t)\mathbf{N}(t) + 0\mathbf{B}(t)$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



$$\mathbf{N}'(t) \cdot \mathbf{B}(t) = ?$$

$$\mathbf{N}(t) \cdot \mathbf{B}(t) = 0$$

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$$\mathbf{B}(t) \cdot \mathbf{T}(t) = 0$$

$$\mathbf{B}'(t) \cdot \mathbf{T}(t) + \mathbf{B}(t) \cdot \mathbf{T}'(t) = 0$$

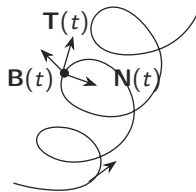
$$\mathbf{B}'(t) \cdot \mathbf{T}(t) + \mathbf{B}(t) \cdot \kappa(t)\mathbf{N}(t) = 0$$

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Exercise: Show that if the torsion of a curve is constant 0, then the curve lies on a plane

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



$$N'(t) \cdot B(t) = ?$$

$$N(t) \cdot B(t) = 0$$

$$N'(t) \cdot B(t) = -N(t) \cdot B'(t) = \tau(t) \text{ (Torsion of } \theta \text{ at } \theta(t))$$

$$B(t) \cdot T(t) = 0$$

$$B'(t) \cdot T(t) + B(t) \cdot T'(t) = 0$$

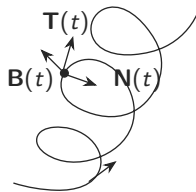
$$B'(t) \cdot T(t) + B(t) \cdot \kappa(t)N(t) = 0$$

$$B'(t) \cdot T(t) = 0$$

Exercise: Show that if the torsion of a curve is constant 0, then the curve lies on a plane

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



$$\mathbf{N}'(t) \cdot \mathbf{B}(t) = ?$$

$$\mathbf{N}(t) \cdot \mathbf{B}(t) = 0$$

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$$\mathbf{B}(t) \cdot \mathbf{T}(t) = 0$$

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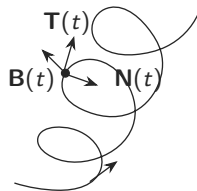
$$\mathbf{B}'(t) \cdot \mathbf{T}(t) + \mathbf{B}(t) \cdot \kappa(t) \mathbf{N}(t) = 0$$

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Exercise: Show that if the torsion of a curve is constant 0, then the curve lies on a plane

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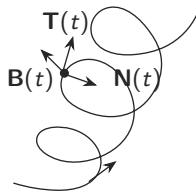
$$\mathbf{B}'(t) \cdot \mathbf{T}(t) = 0$$

Exercise: Show that if the torsion of a curve is constant 0, then the curve lies on a plane

Surfaces
 $U \subset \mathbb{R}^2$

Example

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Exercise: Show that if the torsion of a curve is constant 0, then the curve lies on a plane

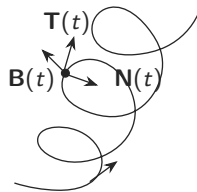
Surfaces

$$U \subset \mathbb{R}^2$$

$$\sigma : U \rightarrow \mathbb{R}^3$$

Example

$$\theta(t) = (\cos(t), \sin(t), t)$$



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Surfaces

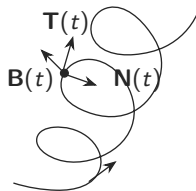
$$U \subset \mathbb{R}^2$$

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Surfaces

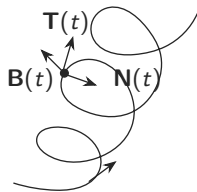
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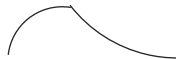


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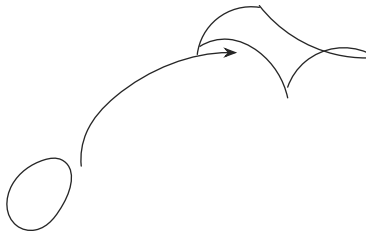


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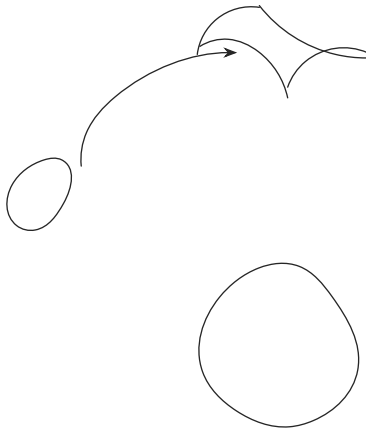


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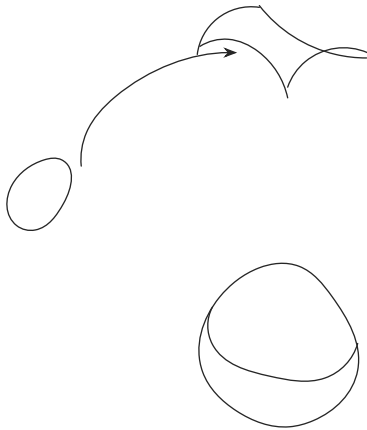


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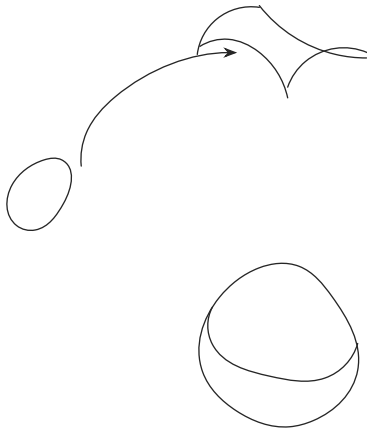


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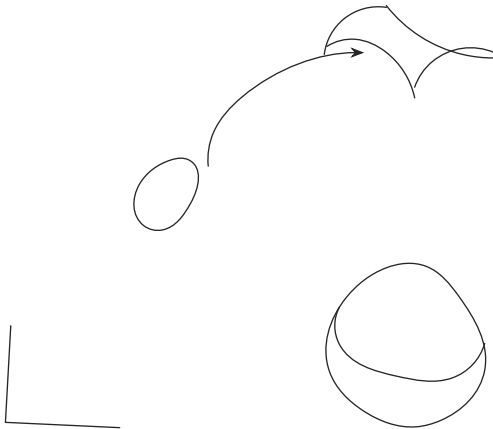


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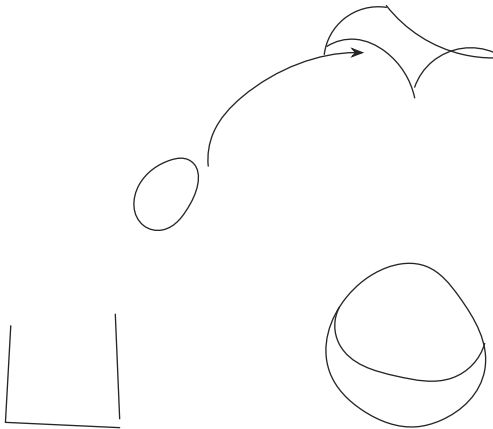


Surfaces

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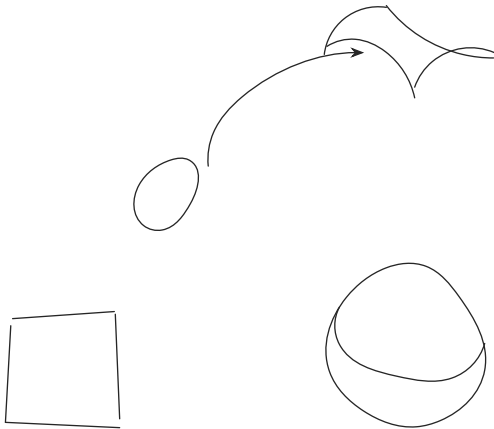


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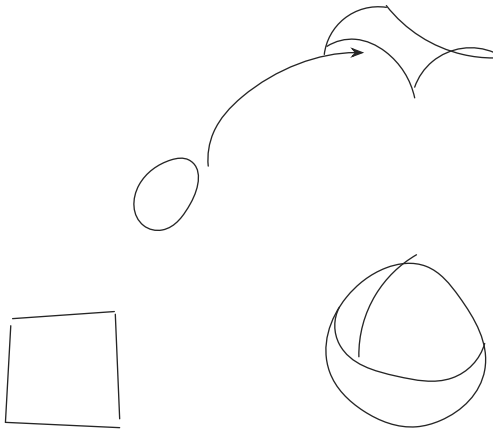


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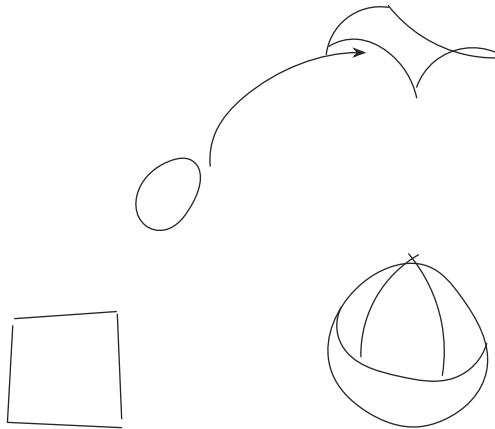


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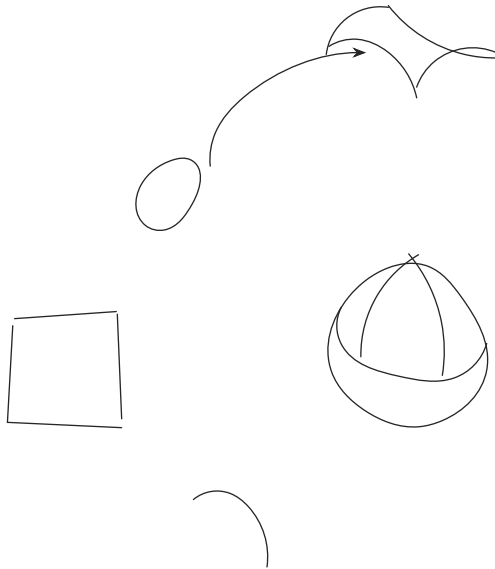


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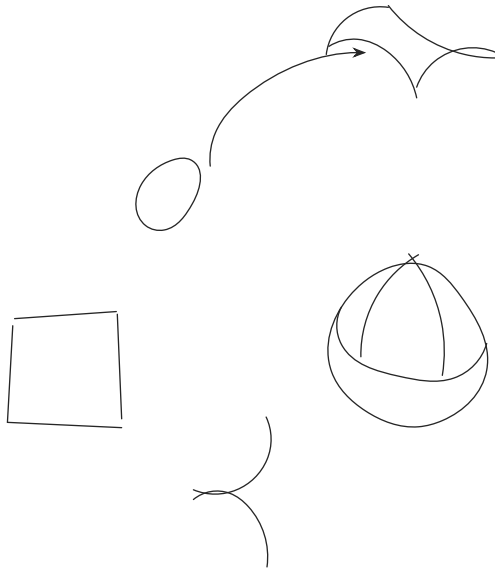


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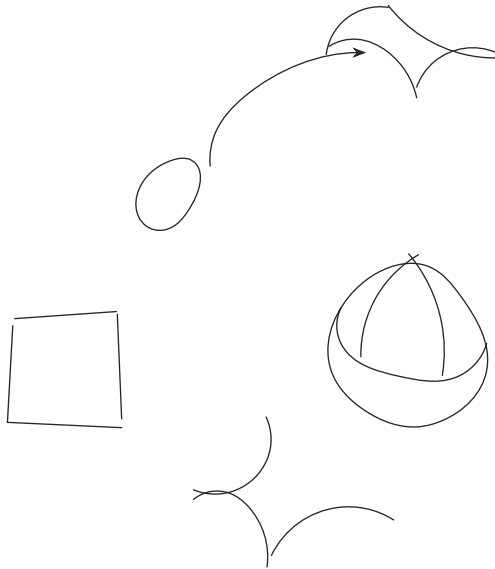


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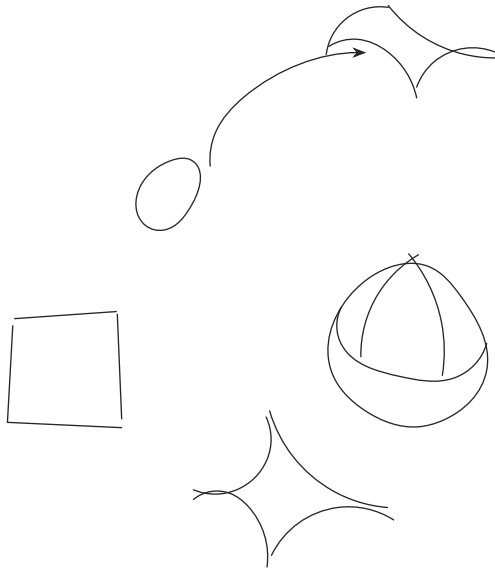


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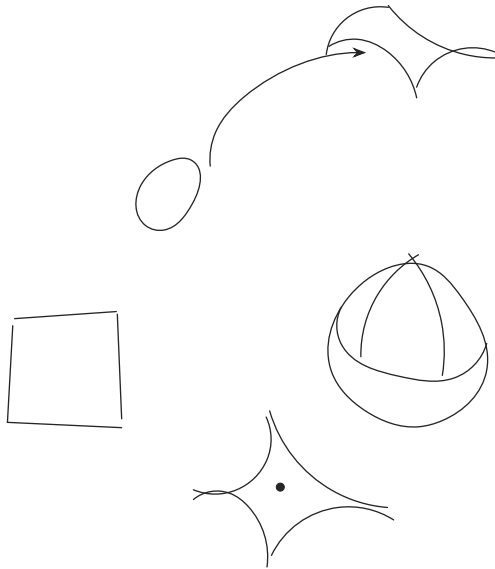


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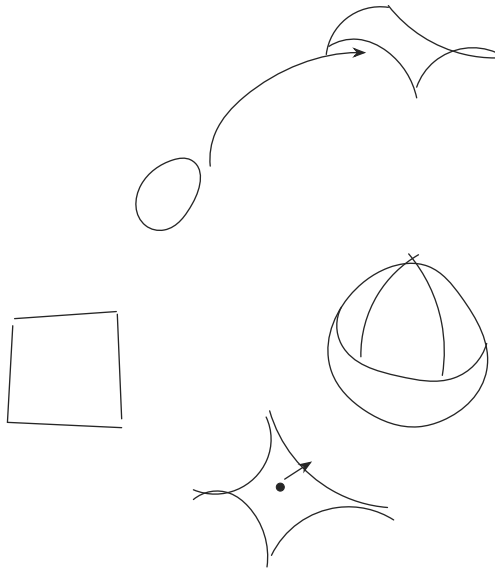


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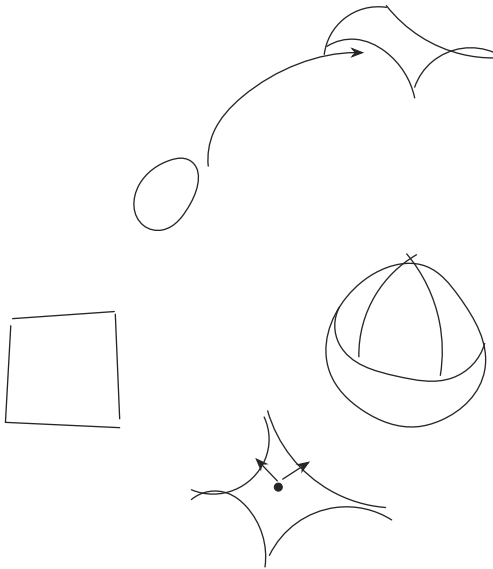


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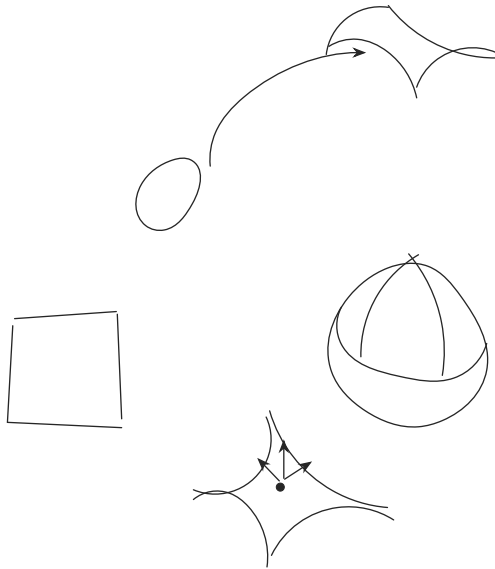


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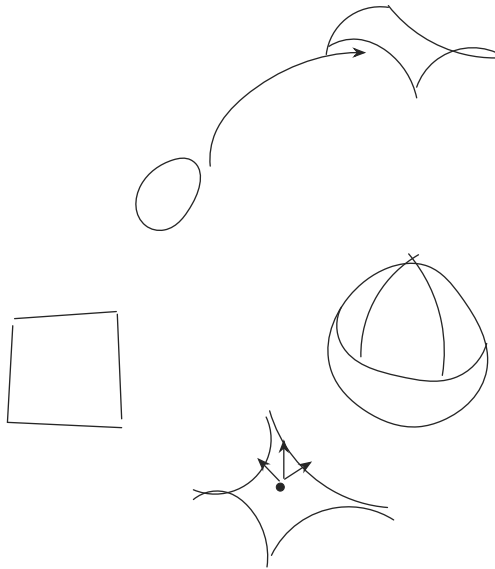


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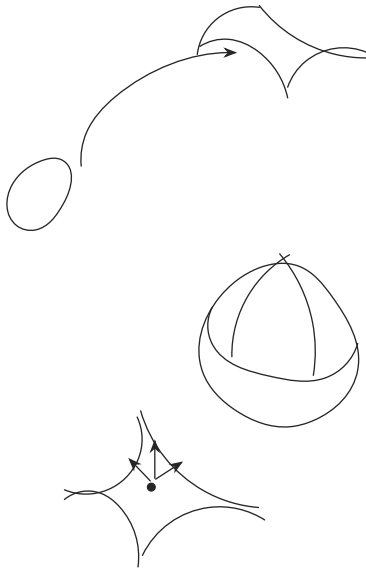
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Look for “Theorem Egregium”



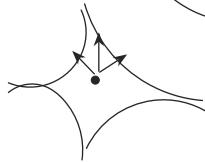
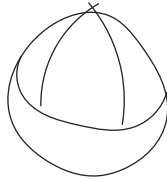
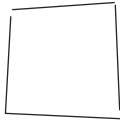
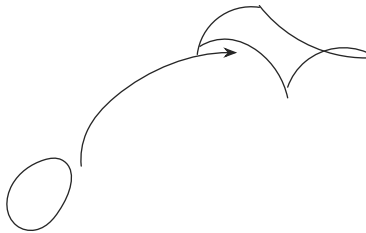
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Look for “Theorem Egregium”
and “Gaussian curvature”



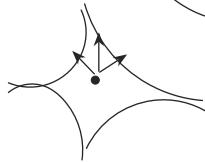
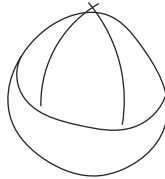
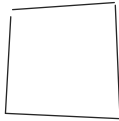
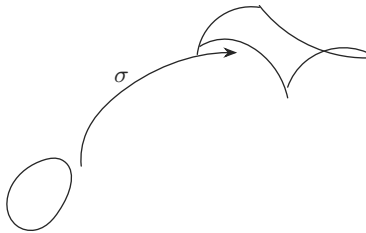
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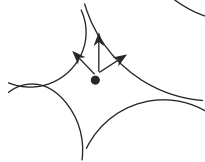
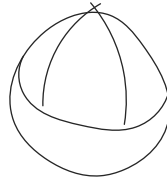
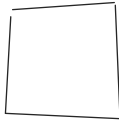
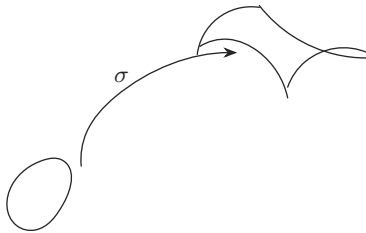
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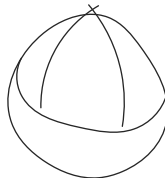
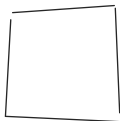
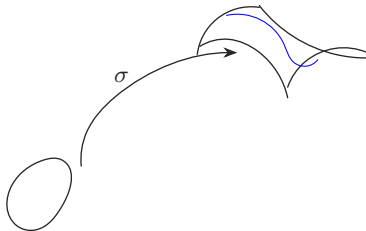
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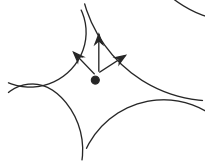
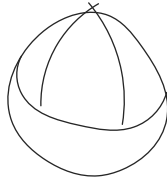
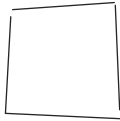
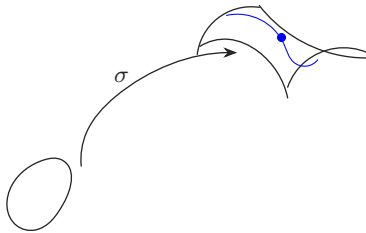
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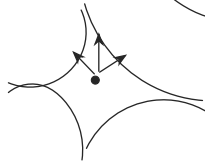
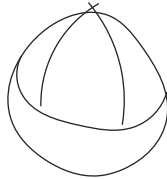
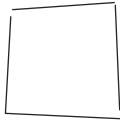
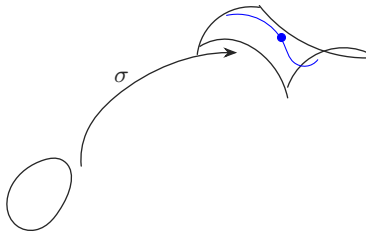
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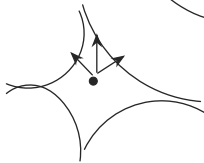
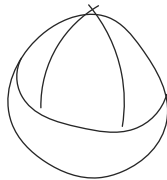
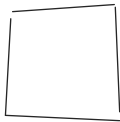
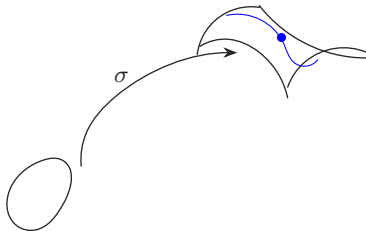
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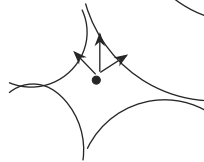
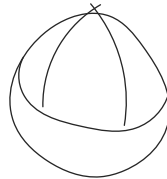
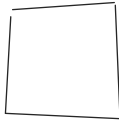
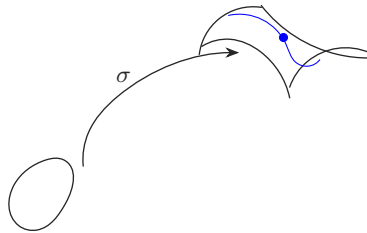
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Look for “Theorem Egregium”
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When moving on a curved surface, there must be an
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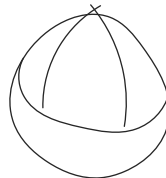
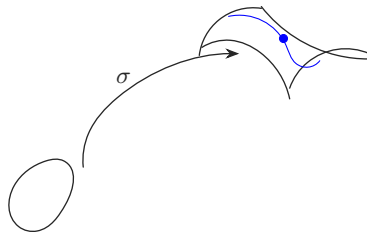
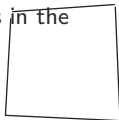
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Look for “Theorem Egregium”
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When moving on a curved surface, there must be an
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The acceleration that keeps it on the surface is in the
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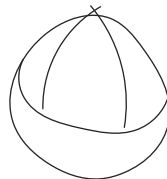
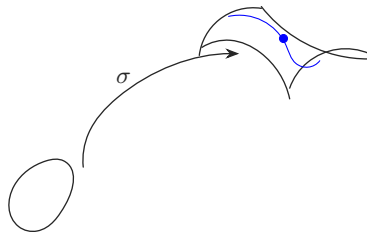
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Look for “Theorem Egregium”
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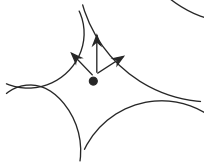
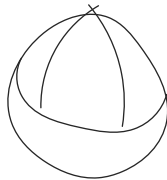
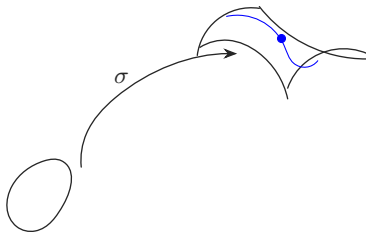
Look for “Theorem Egregium”
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When moving on a curved surface, there must be an acceleration which keeps the point on the surface

The acceleration that keeps it on the surface is in the normal direction

Curvature of a curve on a surface splits into two kinds:

1. The normal curvature which keeps the particle on the surface because the surface is curved. This is forced.
2. The geodesic curvature which is left over curvature.



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